ST MARY'S DSG, KLOOF

MATHEMATICS

GRADE 10 **NOVEMBER 2018**

TIME: 2 HOURS **TOTAL: 122 MARKS**

NAME: Memo **MATHS TEACHER:**_____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of 16 pages and 16 questions.
2. Formulae have been provided below.
3. Write your name on each sheet of paper.
4. Answer all the questions.
5. Diagrams are not necessarily drawn to scale.
6. **All the necessary working details must be clearly shown.**
7. **Reasons must be given unless specified otherwise.**
8. Approved non-programmable calculators may be used except where otherwise stated. Ensure that your calculator is in degree mode.
9. It is in your own interest to write legibly and to present your work neatly.
10. Answers may be left in surd format or rounded off to **two decimal places**.

FORMULAE

$$y = mx + c \qquad m = \frac{y_2 - y_1}{x_2 - x_1} \qquad d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \qquad m = \tan\theta$$

$$M = \left(\frac{x_1 + x_2}{2} ; \frac{y_1 + y_2}{2}\right) \qquad \tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \sin^2\theta + \cos^2\theta = 1$$

SECTION A

[60]
57

Question 1 [14]

The scores in a Mathematics quiz were recorded. The following ordered data of the 40 scores are provided.

13	15	15	15	16	16	16	17	17	18
18	19	19	19	20	20	20	20	20	20
21	21	21	22	22	22	22	23	23	23
25	25	25	26	26	26	27	27	28	28

a) Use the scores above to create a 5-number summary. Fill in the missing headings and values for the five-number summary. (6)

Minimum	Q1	Median	Q3	maximum.
13	18	20, 5	24	28

b) Mrs van Rooyen states that 40% of the scores were less than 20. Is she correct? Justify your answer with an appropriate calculation. (3)

NO $\frac{14}{40}$ are less than 20 35 %

or 40 % = 16 data pts (includes scores of 20)

c) The data above is recorded in a frequency table: Complete the table and then, using the frequency table, calculate an estimated mean. (5)

Score	Midpoint	Frequency
$10 < x \leq 15$	12,5	4
$15 < x \leq 20$	17,5	16
$20 < x \leq 25$	22,5	13
$25 < x \leq 30$	27,5.	7

$(4 \times 12,5 + 16 \times 17,5 + 13 \times 22,5 + 7 \times 27,5) \div 40$

$= 20,38.$

Question 2 **[13]**

a) Find the equation of a straight line through the point $(3\,;\,-2)$ and parallel to the line $9x - 3y + 3 = 0$ (5)

$-3y = -9x - 3$

$y = 3x + 1$

$m = 3 \quad ; \quad y = 3x + c.$

Sub $(3\,;\,-2)$ $\quad : -2 = 9 + c$

$-11 = c$

$y = 3x - 11$

b) $A\,(x\,;\,-2)$, $B(-1\,;\,1)$ and $C\,(5\,;\,10)$ are collinear points.

1. Show that $x = -3$ (4)

$m_{BC} = \frac{9}{6} = \frac{3}{2}$

$m_{AB} = \frac{3}{-1-x} = \frac{3}{2}$

$-1 - x = 2$

$x = -3.$

2. Determine the angle of inclination of line ABC (2)

$m = \frac{3}{2} \quad \tan\theta = \frac{3}{2} \quad \theta = 56,31^{\circ}$

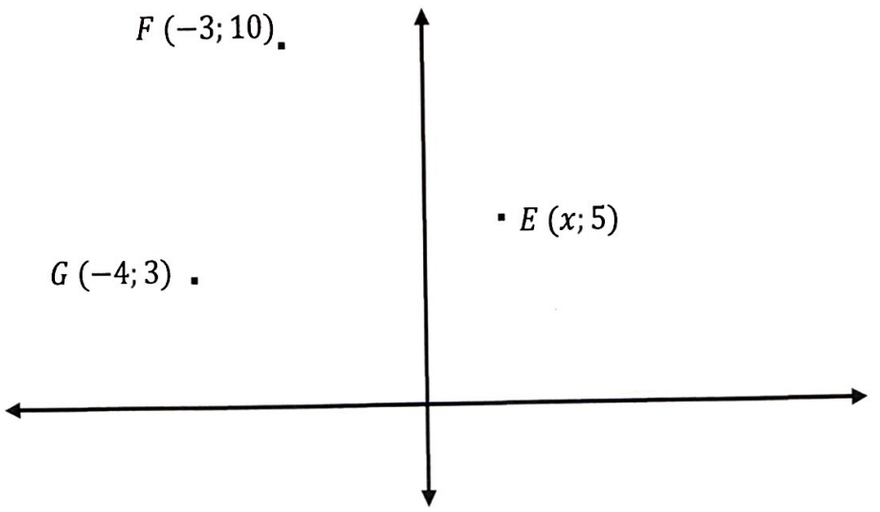
3. If line ABC is the diameter of a circle, determine the co-ordinates of the centre of the circle (2)

$m_{AC} = \left(\frac{-3+5}{2}\,;\,\frac{-2+10}{2}\right) = (1\,;\,4)$

[8]

Question 3

The co-ordinates $E\ (x; 5)$, $F\ (-3; 10)$ and $G\ (-4; 3)$ are shown on the Cartesian plane below.



a) If EF and FG are two sides of a rhombus, ~~show that $x = 2$~~. determine the value of x. (6)

$EF = FG$

$\sqrt{(5-10)^2 + (x+3)^2} = \sqrt{7^2 + 1^2}$

$25 + x^2 + 6x + 9 = 50$

$x^2 + 6x - 16 = 0$

$x = -8$ or $x = 2$

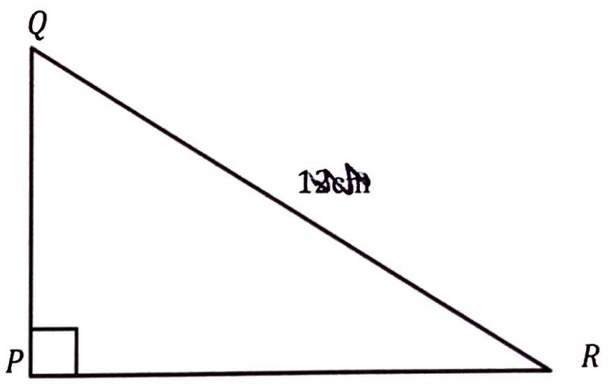
b) Determine the co-ordinates of point H, the 4th vertex of rhombus $EFGH$ (2)

H: $(1; -2)$

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Question 4 [8] 4

Triangle PQR is a right angled triangle with $QR = 12\ cm$ and $\cos\hat{R} = \frac{2}{3}$



a) Determine the length of PR (2)

$\cos\hat{R} = \frac{PR}{QR} = \frac{PR}{12} = \frac{2}{3}$ $\therefore PR = 8.$

b) Determine the length of PQ (2)

$PQ = \sqrt{12^2 - 8^2} = \sqrt{80} = 4\sqrt{5}$

c) Determine the following trig ratios without solving for $\hat{R}$ (2)

1. $\tan\hat{R}$

$= \frac{4\sqrt{5}}{8} = \frac{\sqrt{5}}{2}$

2. $\sin(90° - \hat{R})$ (2)

$= \frac{8}{12} = \frac{2}{3}.$

Question 5 **[10]**

a) Use your calculator to evaluate:

$$\cos\left(\frac{115°}{2}\right) + \frac{\cos 115°}{2}$$ (2)

0,33.

b) Solve for β in the following, round off to ONE decimal digit, β ϵ [0°; 90°]

1. 3 tan β = 6,243 (2)

$\tan \beta = 2,081$

$\beta = 64,33°$

2. cos(β+10°) = 0,13 (2)

$\beta + 10° = 82,53°$

$\beta = 72,53°$

c) Complete the following statements:

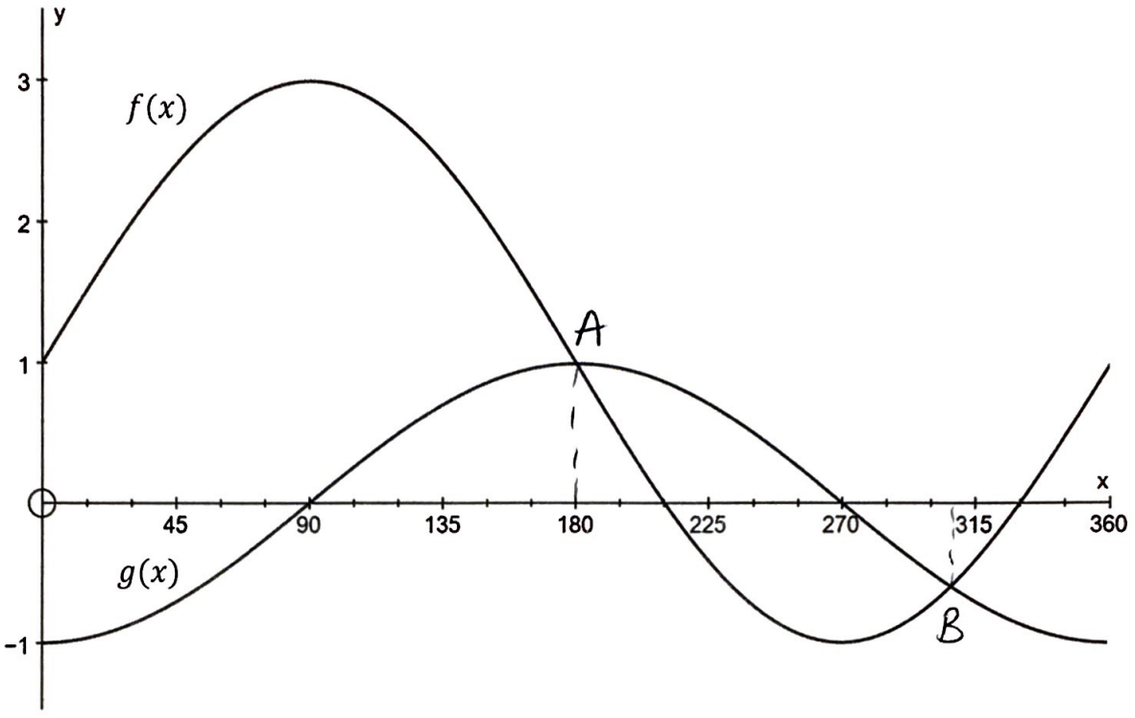
1. If $\cos\theta > 0$ and $\tan\theta < 0$, then $270° < \theta < 360°$ (2)
2. If $\cos\theta . sin\theta > 0$ then θ is in quadrant(s) 1 or 3. (2)

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SECTION B **[63]**

Question 7 **[13]**

The diagram below represents the graphs of the functions $f(x) = a\,sinx + q$ and $g(x) = b\,cosx$



a) Write down the values of a, b *and* q. (3)

$a = 2; \quad q = 1; \quad b = -1$

b) What is the range of f (2)

$y \in [-1; 3]$

c) For which value(s) of x is $f(x)$ decreasing? (2)

$x \in (90°; 270°)$

d) Show on the graph using the letters A,B etc. where you would read off the value(s) of x for which $f(x) = g(x)$. (2)

e) For which value(s) of x is $f(x).g(x) > 0$ (4)

$x \in (90°; 210°) \; ; \; (270°; 330°)$

Question 8 **[7]**

a) Using special angles, show that $1 + \cos 60° = sin^2 60° + \sin 30°$ (show all working) (3)

LHS $= 1 + \left(\frac{1}{2}\right)^2 = \frac{5}{4}$ RHS $= \left(\frac{\sqrt{3}}{2}\right)^2 + \frac{1}{2}$

$= \frac{3}{4} + \frac{2}{4} = \frac{5}{4}$

b) Using trigonometric identities, show that

$$\frac{\sin\theta \cos^2\theta}{\cos\theta - \sin^2\theta \cos\theta} = \tan\theta$$ (4)

LHS: $\frac{\sin\theta \cos^2\theta}{\cos\theta(1 - \sin^2\theta)}$

$= \frac{\sin\theta \cos^2\theta}{\cos\theta \cdot \cos^2\theta}$

$= \frac{\sin\theta}{\cos\theta} = \tan\theta = RHS$

Question 9 **[6]**

A data set of gymnasts' final scores in a competition has the following statistical attributes:

- Mean: 76%
- Mode: 83%
- Maximum: 94%
- Lower quartile: 51%
- Minimum: 31%
- IQR: 37%

Determine whether each statement below is true or false and tick the appropriate column.

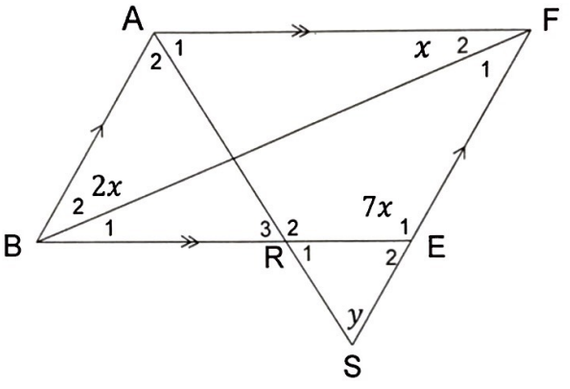
	True	False
The range of the data is 63%.	✓	
The top quartile of gymnasts were within 6% of each other's scores.	✓	
There were more scores below the median than above it.		✓
The most common score lay between 51% and 76%.		✓
Every gymnast scored more than 37%.		✓
The 75th percentile of the scores is 88%.	✓	

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All geometry questions from this point on require reasons

Question 10 **[8]**

$ABEF$ is a parallelogram. $\hat{B}_2 = 2x$, $\hat{E}_1 = 7x$ and $\hat{F}_2 = x$. $\hat{S} = y$.



a) Determine the value of $\hat{F}_1$ in terms of x. (2)

$\hat{F}_1 = 2x$ (alt ∠s AB // EF)

b) Hence, determine the value of x. (3)

$x + 2x + 7x = 180°$ (co-int ∠s AF // BE)

$10x = 180°$

$x = 18°$

c) If $AS = FS$, determine the value of y. (3)

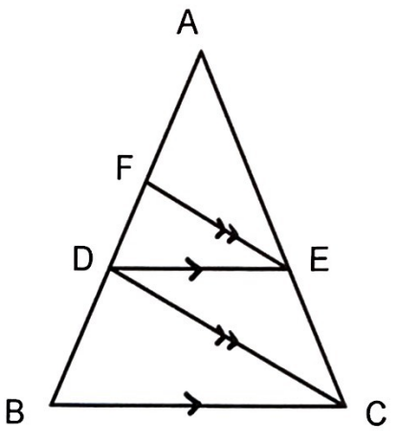
$\hat{A}_1 = \hat{F}_{1+2} = 3x = 3(18°) = 54°$ (∠s opp = sides)

$y = 180° - 2(54°)$ (sum ∠ Δ)

$= 72°$

Question 11 **[8]**

Given triangle ABC with BC // DE,
EF // CD, AF = 80m and FD = 120m.



a) Prove that Δ ADE ||| Δ ABC (3)

In Δ ADE and Δ ABC
A is common
$\hat{ADE} = \hat{ABC}$ (corresp ∠s DE // BC)
$\hat{AED} = \hat{ACB}$ (sum ∠ Δ)
∴ Δ ADE ||| Δ ABC (AAA)

b) Calculate AB giving reasons (Note: diagram is not to scale) (5)

$\frac{AF}{FD} = \frac{AE}{EC}$ (prop int FE // DC)

$\frac{80x}{120x} = \frac{2}{3} = \frac{AE}{EC}$

$\frac{AD}{AB} = \frac{AE}{AC}$ (prop int DE // BC)

$\frac{200}{AB} = \frac{2y}{5y}$ AB = 500 mm

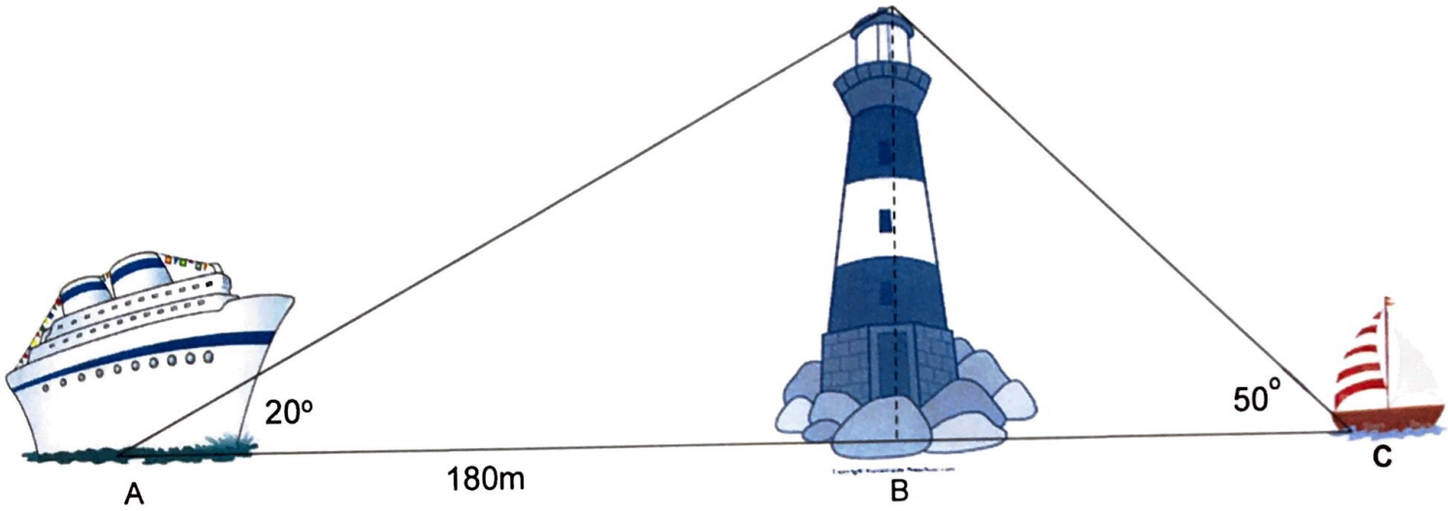
Question 13 [5]

A cruise liner at position A is 120m away from the lighthouse at position B. The angle of elevation from the cruise liner to the top of lighthouse is 20°

. The angle of elevation from a sailboat at position C to the top of the litghthouse is 50°

.

Determine the distance between the cruise liner and the sailboat (AC).



In Δ ALB $\tan 20° = \frac{LB}{180}$ $LB = 65,51m$.

In Δ LBC $\tan 50° = \frac{LB}{BC} = \frac{65,51}{BC}$

$BC = 54,97m.$

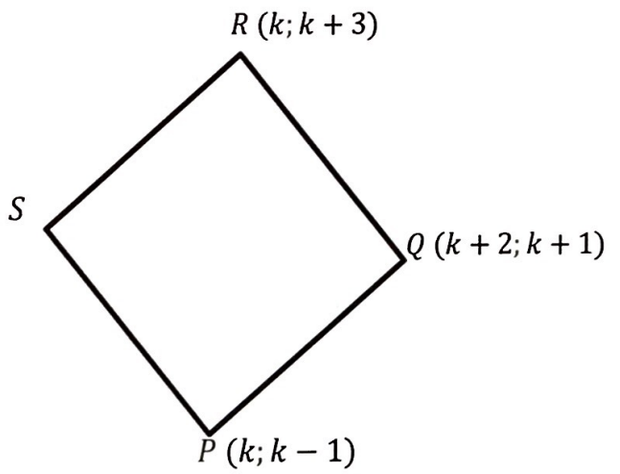
$AC = 180 + 54,97$

$= 234,97\ m$

Question 14 **[5]**

Given quadrilateral $PQRS$ where all 4 sides are equal in length.

Is $PQRS$ a rhombus or a square? Show all supporting working.



$m_{PQ} = \frac{k+1-(k-1)}{k+2-k} = \frac{2}{2} = 1$

$m_{QR} = \frac{k+3-(k+1)}{k-(k+2)} = \frac{2}{-2} = -1$

$\therefore \hat{Q} = 90° \therefore$ PQRS is rhombus

Question 15 [4]

In the diagram, rectangle PQRS has side PQ on the diameter of the semicircle, with R and S on the semicircle. If the diameter of the semicircle is 20 cm and the length of PQ is 16, determine the length of PS. (4)

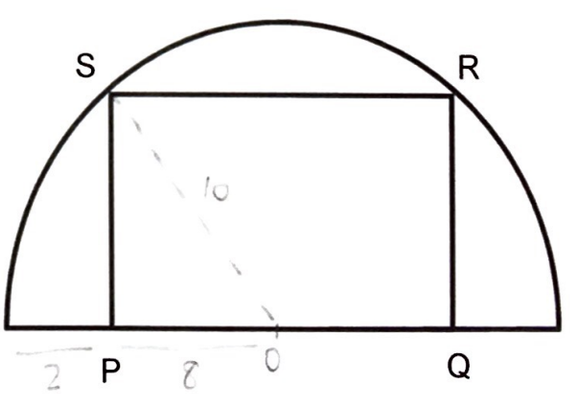
Let O be centre.

$OP = 8$

$OS = 10$ (radii)

$PS = \sqrt{10^2 - 8^2}$

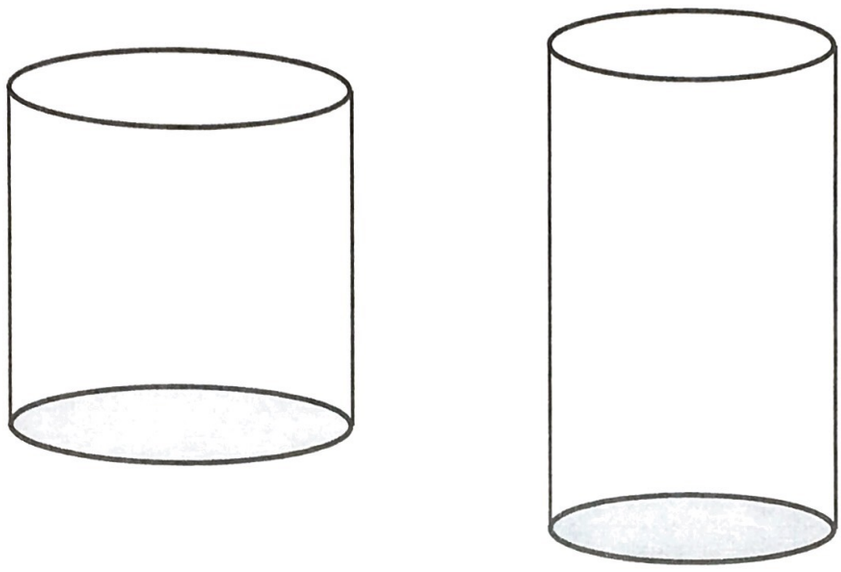
$= \sqrt{36} = 6$



Question 16 **[5]**

A soft drink manufacturer has been packaging their drinks into cylindrical cans with a diameter of 6cm and a height of 9cm.

As part of a new innovation, they consider changing the dimensions of the can. If the base of the can is reduced to a diameter of 5cm, how much taller will the new can have to be in order to hold the same volume?



$\text{A} = \left(\frac{5}{6}\right)^2 = \frac{25}{36}$ or $V_{original} = \pi r^2 h$

$= \pi(3)^2(9)$

$= 81\pi$

$9 \times \frac{36}{25} = 12,96$

$V_{new} = 81\pi = \pi(2,5)^2 h$

$h = 12,96$

$\therefore 12,96 - 9 = 3,96$ cm taller.